FPT SCHOOL OF BUSINESS & TECHNOLOGY

# SEMANTIC LANE AND TRAFFIC SIGN UNDERSTANDING USING VISION-LANGUAGE MODELS FOR DRIVING DECISION SUPPORT

Academic Supervisor: **Dr. Doan Nhat Quang**
Student: **Do Minh Hieu - MSE13172**

Hanoi, 25/10/2026

# Outline

FPT SCHOOL OF BUSINESS & TECHNOLOGY


1 Introduction

2 Literature Review and Theoretical Background

3 Proposed Methodology

4 Experiments and Results

5 Conclusion

FPT SCHOOL OF BUSINESS & TECHNOLOGY

# Introduction

Motivation · Research gap · Problem · Objectives · Contributions

# Background and Motivation

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## Context

- **ADAS:** lane-departure warning, lane keeping, traffic sign recognition
- Lane-perception functions substantially reduce crash and injury risk (IIHS)
- **VLMs:** image + text → situation description and recommendations in natural language

## Limitations of VLMs

- **Weak spatial reasoning:** which lane, how far off-center
- **Hallucination:** claims without visual evidence
- **Remedy:** add structured information (RAG, DriveVLM, Talk2BEV)

→ Can semantics extracted from detection models help VLMs give better driving recommendations?

# Research Gap and Problem Statement

FPT SCHOOL OF BUSINESS & TECHNOLOGY




- **Gap:** the role of semantics built from detection outputs in VLM recommendation quality has not been verified in isolation
- **Problem:**
  - Convert lane and sign detection outputs → Structured Semantic Information (SSI)
  - Use SSI as context for the VLM to generate driving recommendations
  - Evaluate by controlled comparison: same VLM, same data

→ **Research question:** Does adding SSI change recommendation quality compared with image-only input, and by how much?

# Research Objectives

FPT SCHOOL OF BUSINESS & TECHNOLOGY

1

## Understanding traffic semantics

Module that extracts SSI from lane and sign detection results

2

## SSI-based recommendation generation

Feed SSI together with the image to the VLM via a prompt

3

## Evaluating the role of SSI

Quantify the change in recommendation quality when SSI is added

# Contributions of the Thesis

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## 1 Semantic analysis and recommendation pipeline

- SSI from deterministic geometric rules: no extra model, no learnable parameters
- SSI + image → VLM via a prompt with evidence-use and conflict-resolution rules; no fine-tuning

## 2 Controlled evaluation framework for the role of SSI

- With vs. without SSI on the same VLM and the same data
- Multi-criterion rubric, multiple LLM judges, human comparison, statistical testing
- Analysis by SSI accuracy

FPT SCHOOL OF BUSINESS & TECHNOLOGY

# Literature Review

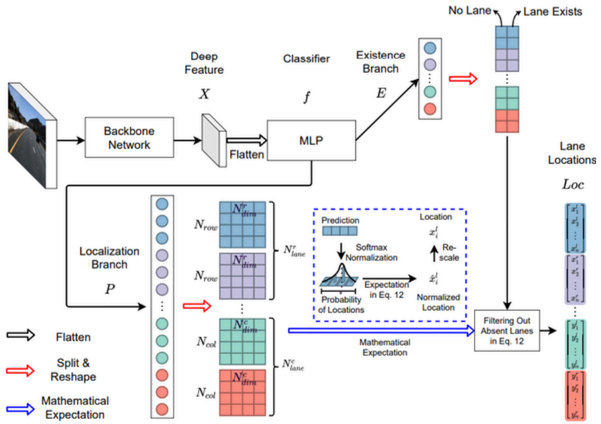
Lane and sign detection · VLMs in traffic · LLM-as-a-Judge

# Lane and Traffic Sign Detection

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## Lane detection

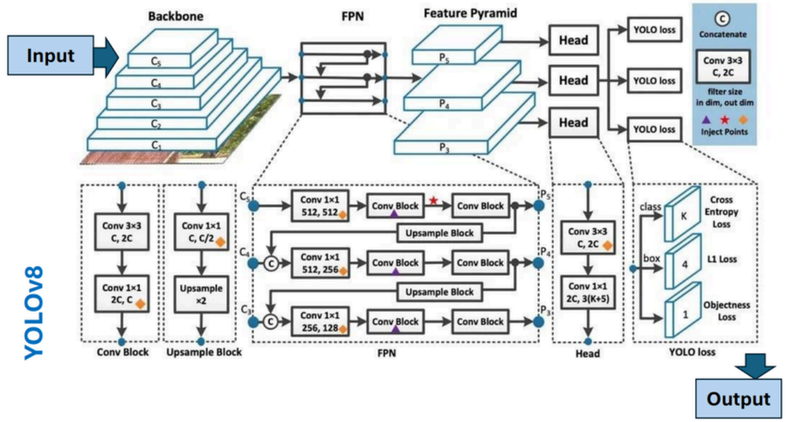
Segmentation · row/column classification (UFLD-v2) · anchor-based (LaneATT, CLRNet)




- **UFLD-v2:** Existence + Localization branches → point coordinates on lane boundaries

## Traffic sign detection

Two-stage (Mask R-CNN) · one-stage (YOLO)




- **YOLOv8:** C2f backbone, 3-level FPN, anchor-free head, NMS

→ Detection output is the input to the semantic step: needs boundary point coordinates, sign class and position

# VLMs for Traffic Applications

FPT SCHOOL OF BUSINESS & TECHNOLOGY

| Group | Representative works | Approach | Mismatch with this thesis |
|---|---|---|---|
| End-to-end | DriveGPT4, DriveLM, LMDrive | Language inside the control loop | Needs large driving data and training infrastructure |
| Reasoning layer on an existing pipeline | **DriveVLM, Talk2BEV (closest)** | Structured scene representation for the VLM | 3D/BEV, multi-sensor; contribution of the representation not quantified separately |
| CNN + MLLM fusion | SafeRoute | CNN features fused into the MLLM | Requires MLLM fine-tuning |

→ **This thesis:** 2D representation from lightweight detectors, single camera, no fine-tuning, **directly measures the representation's contribution**

# Output Quality Evaluation with LLM-as-a-Judge

FPT SCHOOL OF BUSINESS & TECHNOLOGY

- **Why:** free-text recommendations, no single correct answer → accuracy, F1 not applicable
- **Method:** an LLM scores outputs against a rubric (MT-Bench, Chatbot Arena, AlpacaEval)
- **Risks:** length, style and model biases → judge scores are not ground truth
- **Recommendations:** add human evaluation (Kim); task-specific criteria (Pan et al.)

**85%**

GPT-4 vs. human experts

**81%**

expert vs. expert

*Agreement on MT-Bench (Zheng et al.)*

→ **This thesis:** task-specific rubric · 3 judges · comparison with human raters

FPT SCHOOL OF BUSINESS & TECHNOLOGY

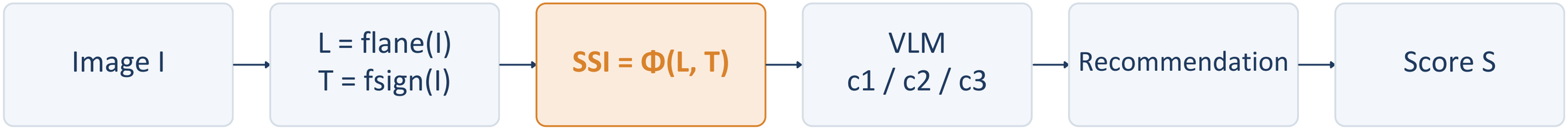
# Proposed Methodology

Problem formulation · Architecture · Model and data selection
SSI extraction · Recommendation generation · Evaluation

# Problem Formulation

FPT SCHOOL OF BUSINESS & TECHNOLOGY

Formalization, assumptions and constraints




## Input modes and score

**c1 = I** Image-only
**c2 = SSI** SSI-only
**c3 = (I, SSI)** Combined
**$S_{i,m} = (1/6) \Sigma_k s_{i,m,k}$**, $s \in \{1,\ldots,5\}$, 6 criteria

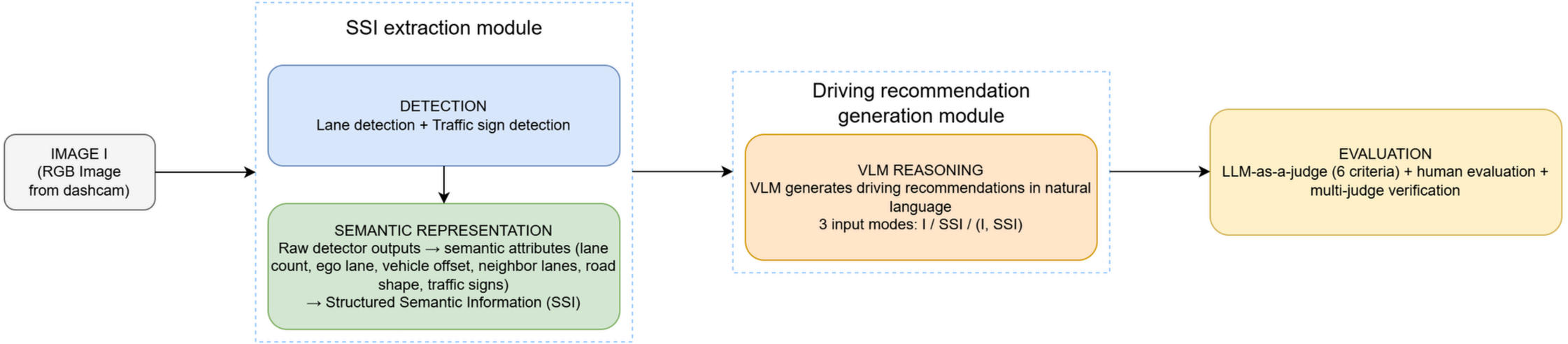
- **Assumptions:**
  - Independent frames · camera at vehicle center
  - No depth estimation · SSI only from detection outputs
- **Constraints:**
  - CPU only · VLM via API, no fine-tuning

→ Main comparison: **c3 vs. c1**, paired per image; c2 as a reference

# Overall System Architecture

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## Pipeline modules




1. **Detection:** UFLD-v2 (lanes) + YOLOv8n (signs)
2. **Semantic representation Φ:** geometric rules → SSI (JSON)
3. **Recommendation generation:** VLM, 3 input modes
4. **Evaluation:** LLM-as-a-Judge + human raters

# Overall System Architecture

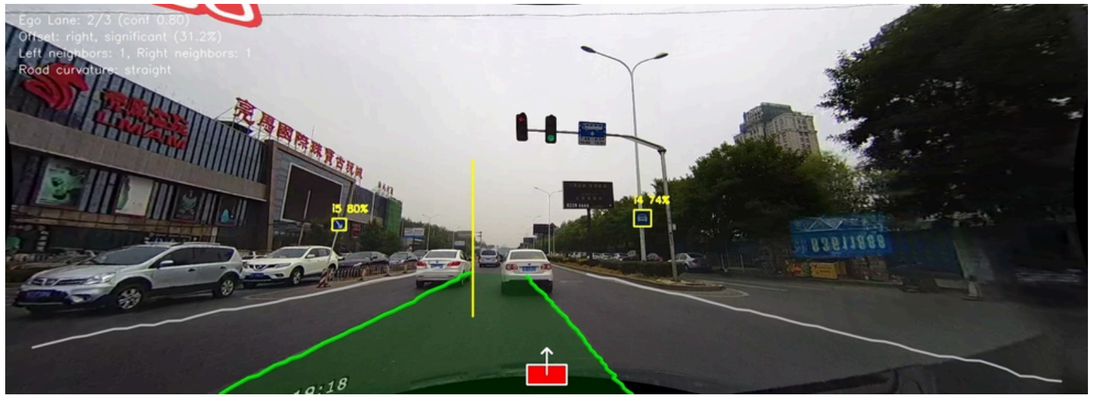
FPT SCHOOL OF BUSINESS & TECHNOLOGY

Example: one image through the system

## 1 Input image



## 2 Detection: lane boundaries + sign boxes




## 3 SSI

**Lane count:** 3
**Ego lane:** 2/3
**Offset:** right, significant, 31.2%
**Road shape:** straight
**Signs:** i5 (left), i4 (center)

## 4 VLM recommendation (Combined mode)

**Situation:** The ego vehicle is in the middle lane (2/3) of a three-lane road, offset 31.2% to the right of the lane center. The road is straight…
**Recommendation:** Maintain the current lane and position…
**Safety note:** …be aware of lateral position to prevent drifting into adjacent lanes.

# Model and Data Selection

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## Lane detection

**Criteria:** correct function · runs on CPU · output suits the next step

| Model (CULane) | Backbone | F1 (%) | FPS* |
|---|---|---|---|
| LaneATT | ResNet-34 | 76.6 | 171 |
| **UFLD-v2** | **ResNet-34** | **76.0** | **156** |
| CLRNet | ResNet-34 | 79.7 | — |
| CLRNet | DLA-34 | 80.5 | — |

| Dataset | Images | Conditions | UFLD-v2 checkpoint |
|---|---|---|---|
| **CULane** | **133,235** | **9 scenario categories** | **Yes** |
| CurveLanes | 150,000 | Curved roads | Yes |
| TuSimple | 6,408 | Highway, daytime | Yes |
| LLAMAS | 100,042 | Highway | No |

→ **Choose UFLD-v2 ResNet-34 pretrained on CULane:** direct point coordinates · no NMS → runs on CPU · most diverse data

**FPS measured on an RTX 3090, for reference only*

# Model and Data Selection

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## Traffic sign detection

| Dataset | Images | Classes | Remark |
|---|---|---|---|
| **TT100K** | **100,000** | **221** | **✓ bbox + class** |
| GTSRB | 51,839 | 43 | ✗ cropped, no location |
| LISA | 6,610 | 47 | ✓ but small |
| BDD100K | 100,000 | 1 | ✗ no classification |
| Mapillary | 105,000 | 400 | ✓ but global scope |

| Model | Params | GFLOPs | mAP50-95 (COCO) | CPU |
|---|---|---|---|---|
| **YOLOv8n** | **3.2M** | **8.7** | **37.3** | **80.4 ms** |
| YOLOv8s | 11.2M | 28.6 | 44.9 | 128.4 ms |
| YOLOv8m | 25.9M | 78.9 | 50.2 | 234.7 ms |
| RT-DETR-R18 | 20M | 60 | 46.5 | — |

→ **Choose YOLOv8n, fine-tuned on TT100K with 50 classes:** CPU speed prioritized over maximum mAP

*Fine-tuning: Kaggle GPU · 640×640 · SGD, lr 0.01 · ≤ 100 epochs, early stopping (patience 30)*

# Model and Data Selection

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## Evaluation data and ground truth

| Dataset | Size | Role |
|---|---|---|
| CULane | 200 random images (176 Normal, 24 Hard) | Main evaluation |
| Self-collected | 200 dashcam frames | Out-of-domain check |

- **Manual ground truth:**
  - Lane count, road shape: labeled independently of the model
  - Ego lane, signs: pipeline output checked against the image
  - Hard: reason recorded (parking garage, intersection, congestion, merging)



(a)



(b)

# Model and Data Selection

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## Candidate VLMs

- **Requirements:** image + text input · pretrained, via API, no fine-tuning
- **NVIDIA NIM platform:**
  - API endpoints for many VLMs, free-access options within the study's scope
  - One platform → unified API-call protocol across models
- **Screening criteria:** execution reliability · output-structure adherence · content quality
  - Same API conditions, same data, Combined mode

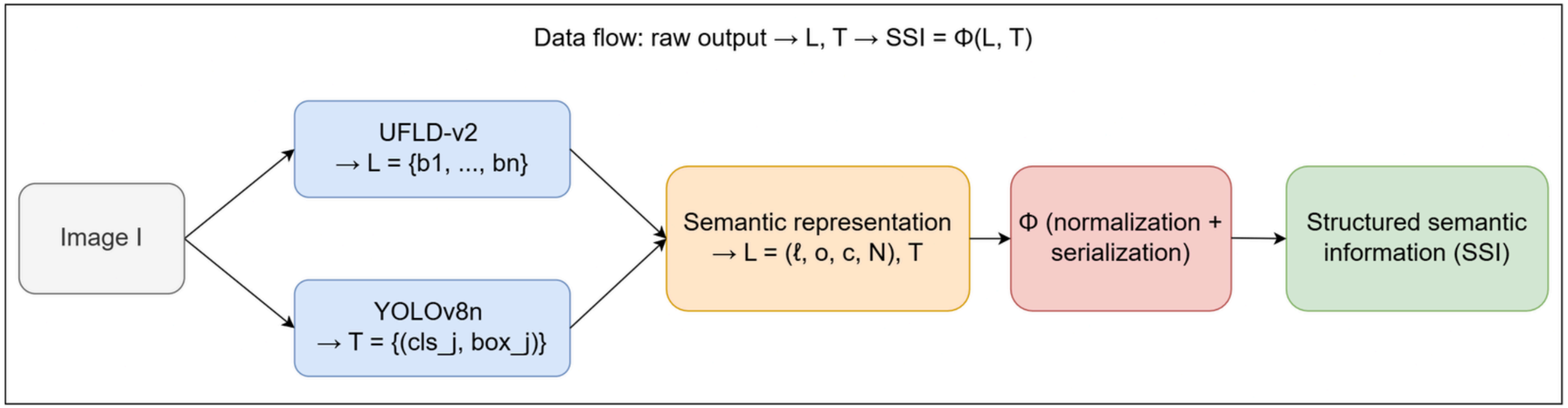
| | |
|---|---|
| **8B** | `nemotron-nano-vl-8b` |
| **12B** | `nemotron-nano-12b-v2-vl` |
| **31B** | `ising-calibration-31b` |

→ The selected model is **kept fixed** for the whole comparative experiment (results in E4)

# Semantic Representation

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## Four design principles




### 1 Driven by recommendation needs

Lane count, ego lane, offset, neighbor lanes, road shape, signs

### 2 Explicit geometric rules

No training data, runs on CPU, explainable

### 3 Meaningful discrete labels

"right, significant, 31.2%" instead of pixel coordinates

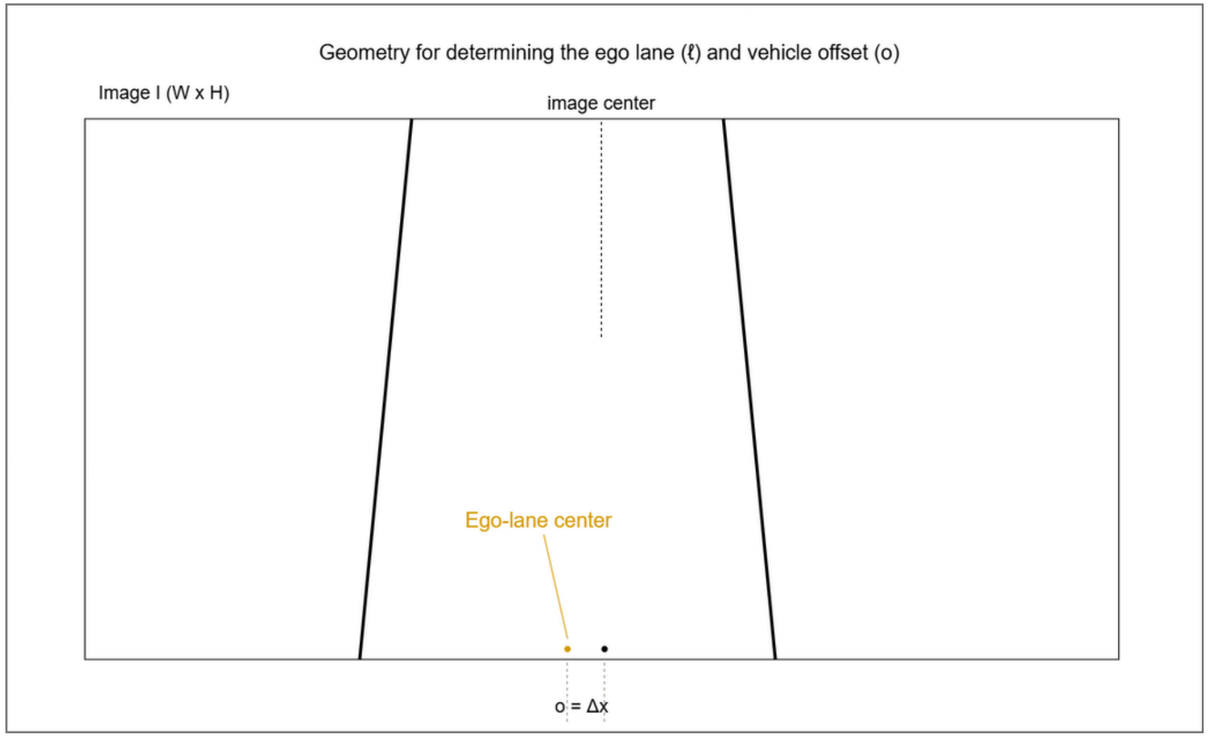
### 4 Verifiable

Each field checked against ground truth or traced to a verified field

# Semantic Representation

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## Lane semantics (1): sorting boundaries and selecting the ego lane




### Step 1 – Sort boundaries

- x at reference row yr = ρ·H (ρ = 0.919, just above the hood)
- Sort left → right

### Step 2 – Select the ego lane

- Each adjacent boundary pair = one candidate lane
- Minimum penalty: close to vehicle center · contains the center · plausible width
- Confidence 0.95 / 0.8 / 0.6 / 0.4 by distance to the center

# Semantic Representation

## Lane semantics (2): offset, road shape, lane count

- **Step 3 – Offset:** Δx% = (x_v − lane center) / lane width × 100
  - < 10%: none · 10–30%: slight · ≥ 30%: significant
- **Step 4 – Curvature:** fit 1st- and 2nd-order polynomials
  - 2nd order clearly better → curved
- **Step 5 – Road shape:** straight · slight curve left/right · sharp curve left/right
- **Step 6 – Lane count:** number of boundaries − 1



straight



gentle curve



sharp curve

*Heuristic thresholds chosen on sample images, not tuned on ground truth*

# Semantic Representation

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## Traffic sign semantics and SSI structure

### Traffic signs

1. Filter detections with confidence ≥ 0.5
2. Class code → sign name, sign type
3. Position: left ($x_c < 0.33W$) · center · right ($x_c > 0.66W$)

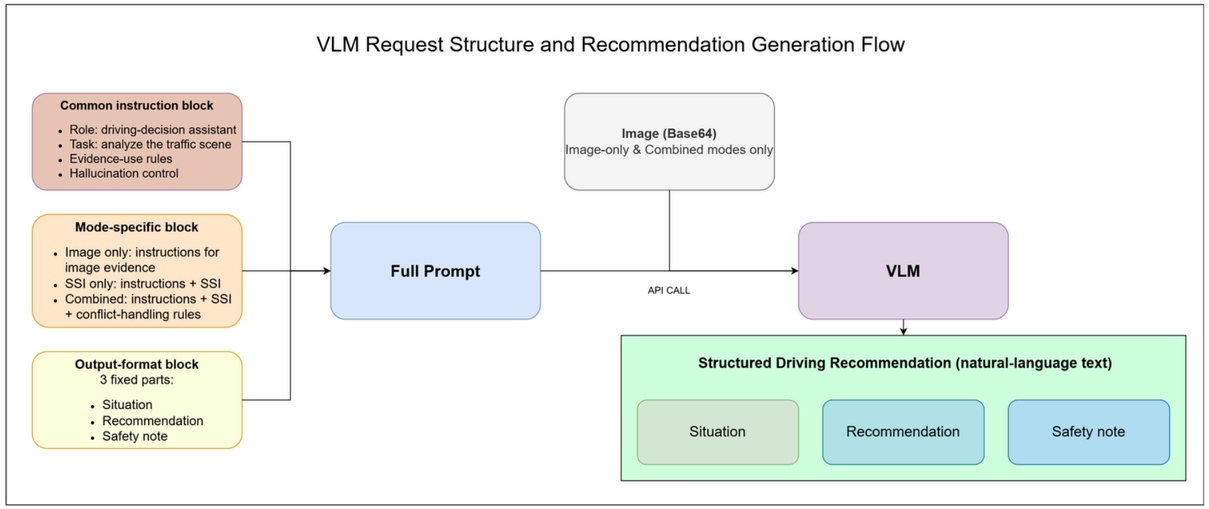
### SSI: one JSON per image, 6 groups


```
{ "lane_count": 3,
  "ego_lane": {"position": "2/3", "confidence": 0.8},
  "vehicle_offset": {"direction": "right",
      "magnitude": "significant", "offset_percent":
31.2},
  "neighbor_lanes": {"left_count": 1, "right_count": 1},
  "road_shape": {"type": "straight", "severity": "none",
      "direction": null},
  "traffic_signs": {"detected": [
    {"sign_code": "i5", "relative_position": "left",
     "sign_name": "Mandatory direction sign"},
    {"sign_code": "i4", "relative_position": "center",
     "sign_name": "Route for motor vehicles only"} ]} }
```

# Recommendation Generation Module

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## Prompt design for three input modes




| Block | Content | Varies by mode |
|---|---|---|
| P_common | Role, task, evidence rules, anti-hallucination | No |
| **B_m** | **Information source and how to use it** | **Yes** |
| P_out | Situation · Recommendation · Safety note | No |

**Pm = Pcommon ⊕ Bm ⊕ Pout**

### Combined mode rules

- Clear image → use the image
- Lanes, offset, signs → use SSI
- Conflict → stronger evidence wins; never assume SSI is correct

*temperature 0.2 · max_tokens 400 · top_p 0.7*

# Evaluation Method

## Experimental design, rubric and rating scale

FPT SCHOOL OF BUSINESS & TECHNOLOGY

| Component | Setting |
|---|---|
| Independent variable | Input mode: Image-only / SSI-only / Combined |
| Dependent variable | 6 criterion scores, aggregate score S |
| Held constant | Images, VLM, common prompt, generation config, judge, rubric |
| Unit of analysis | Image (3 modes of the same image are paired) |

**Situation understanding**

Situation · Road geometry · Ego-lane position · Signs and rules

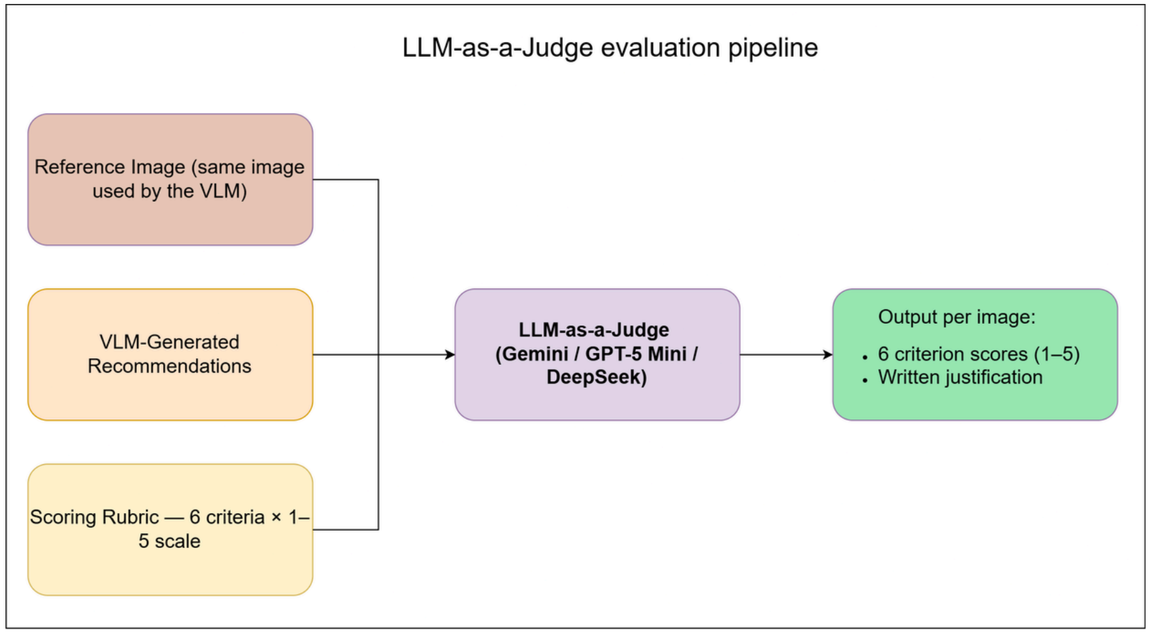
**Decision support**

Driving recommendation · Safety note

| 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|
| correct, well reasoned | minor errors | partly correct | serious errors | wrong / unusable |

# Evaluation Method

## Scoring protocol and statistical testing




- **Per call:** image + 3 mode outputs + rubric → 6 criterion scores with reasons (JSON)
- **Principles:** 3 modes scored together · compare within the same session only · no penalty for information absent from the input
- **Judges:** Gemini (primary), GPT-5 Mini, DeepSeek · human comparison on 20 images

→ **Statistics:** Δ = S(Combined) − S(Image-only) · Wilcoxon signed-rank, Bonferroni (9 comparisons) · Cohen's d_z

FPT SCHOOL OF BUSINESS & TECHNOLOGY

# Experiments and Results

9 experiments: SSI quality · role of SSI · judge reliability
supplementary experiments

# Experimental Setup

FPT SCHOOL OF BUSINESS & TECHNOLOGY

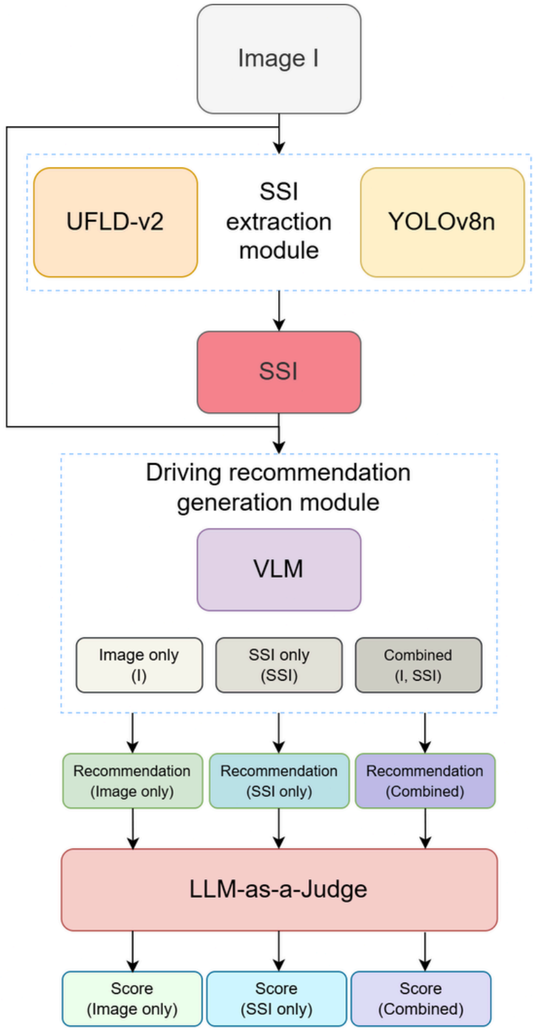
## Experiments and their purpose

| # | Experiment | Question | Serves |
|---|---|---|---|
| E1 | Lane semantics | Are the SSI lane fields correct? | Objective ① |
| E2 | Signs and SSI quality | Are the SSI sign fields correct? | Objective ① |
| E3 | CPU cost | Is SSI lightweight enough? | CPU constraint |
| E4 | VLM screening | Which VLM? | Objective ② |
| **E5** | **Role of SSI** | **How much does SSI improve recommendations?** | **Objective ③ – research question** |
| E6 | SSI accuracy | What if SSI is wrong? | Robustness of Combined mode |
| E7 | Judge vs. human | Can the judges be trusted? | Evaluation reliability |
| E8 | VLM self-extraction | Is the pipeline needed? | Rationale for SSI |
| E9 | Prompt influence | Is the result due to the prompt? | Ruling out a confounder |

# Experimental Setup

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## Environment and models




| | |
|---|---|
| **Hardware** | Ryzen 5 5500U, 16 GB RAM, CPU only |
| **Detection** | UFLD-v2 (CULane) · YOLOv8n (TT100K, confidence 0.5) |
| **VLM** | via NVIDIA NIM, 120 s timeout |
| **Judges** | gemini-3.5-flash · gpt-5-mini · deepseek-v4-flash-vision-exp |

# Experimental Setup

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## Evaluation metrics

### Lane semantics

- Lane Count Accuracy, MAE
- Lane Count Precision
- Ego Lane Accuracy
- Road Shape Accuracy (5 classes)

### Sign detection

- Precision, Recall, F1
- Classification Accuracy

### Judge vs. human

- Exact agreement
- Agreement within ±1 point
- Pearson r

→ **Effect size:** Cohen's $d_z = \bar{d} / s_d$ (0.2 small · 0.5 medium · 0.8 large)

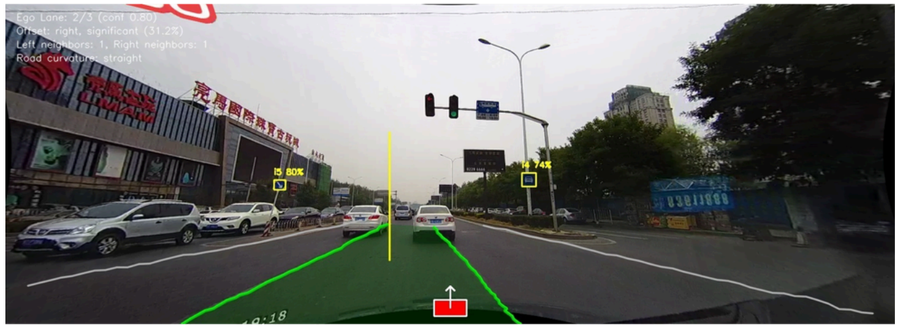
# Evaluation of the SSI Extraction Module

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## E1: lane semantics



(a)




(b)

| Metric | CULane | Normal | Hard | Self-collected |
|---|---|---|---|---|
| **Ego Lane Accuracy** | **86.0%** | **96.6%** | **8.3%** | **86.5%** |
| Lane Count Accuracy | 69.0% | 77.3% | 8.3% | 50.5% |
| Lane Count MAE | 0.475 | 0.273 | 1.958 | 0.715 |
| Lane Count Precision | 96.6% | 96.2% | 100% | 99.4% |
| Road Shape Accuracy | 76.5% | 84.1% | 20.8% | 59.0% |

→ Ego lane stable at ~86% on both datasets · errors are mostly missed lanes, rarely spurious ones · weak on Hard images (19/24 with no lane detected)

# Evaluation of the SSI Extraction Module

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## E2: traffic signs and SSI quality

**YOLOv8n on TT100K val:** P 86.8% · R 67.0% · mAP50 79.8%

| Metric | CULane (20 signs) | Self-collected (72 signs) |
|---|---|---|
| Precision | 84.6% | 82.7% |
| **Recall** | **55.0%** | **59.7%** |
| F1 | 66.7% | 69.4% |
| Classification Acc. | 100% | 82.7% |

### Verification level of SSI fields

**Direct** — lane count, ego lane, road shape, sign class

**Derived** — offset, neighbor lanes

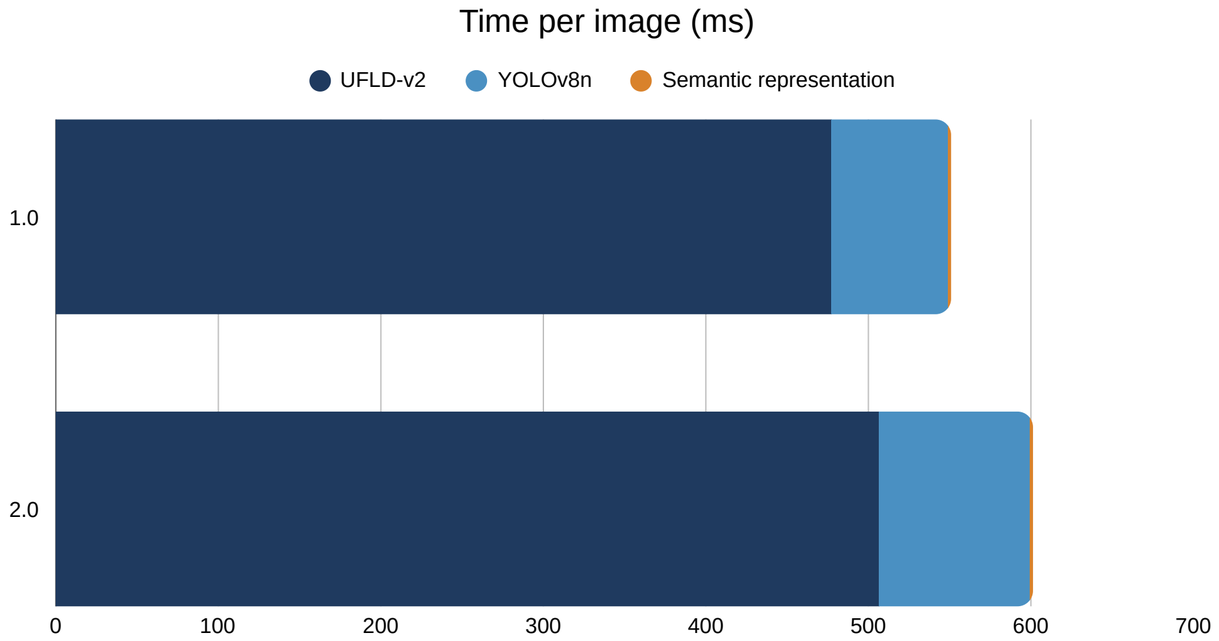
**Heuristic 2D** — sign position

→ Detected signs are mostly classified correctly · weakness: **misses** (recall 55–60%) and signs outside the 50 classes → Combined mode needed

# Evaluation of the SSI Extraction Module

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## E3: computational cost on CPU




| Component (ms/image) | CULane | Self-collected |
|---|---|---|
| UFLD-v2 | 477.5 | 506.6 |
| YOLOv8n | 71.7 | 92.9 |
| **Semantic representation** | **1.7** | **1.8** |
| **Total** | **550.8** | **601.3** |

- YOLOv8s ×2.0–2.2 slower, YOLOv8m ×4.3–4.5 slower than YOLOv8n

→ Semantic step takes only 0.3% of the time · SSI extraction ≈ 12% of the VLM response time (4.52 s)

# Evaluation of Recommendation Quality

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## E4: VLM screening

| | ising-calibration-31b | nemotron-nano-vl-8b | nemotron-nano-12b-v2-vl |
|---|---|---|---|
| Successful responses | 200/200 | 200/200 | 34/200 |
| Mean response time | 4.52 s | 11.75 s | — |
| Outputs < 80 characters | 0% | 49.5% | — |
| 6-criterion score | **3.68** | 2.32 | excluded |



nemotron-nano-vl-8b (Combined)

Maintain your current lane. The road is clear, and there are no signs indicating a need to change lanes.

ising-calibration-31b (Combined)

Situation: The ego-vehicle is currently positioned in the middle lane (2/3) of a three-lane road, with a significant offset to the right side of its lane. The road exhibits a gentle curve, and there are no visible traffic signs or signals in the immediate vicinity.

Recommendation: To improve safety and maintain proper lane discipline, the driver should gradually steer the vehicle to the left to center it within the ego lane.

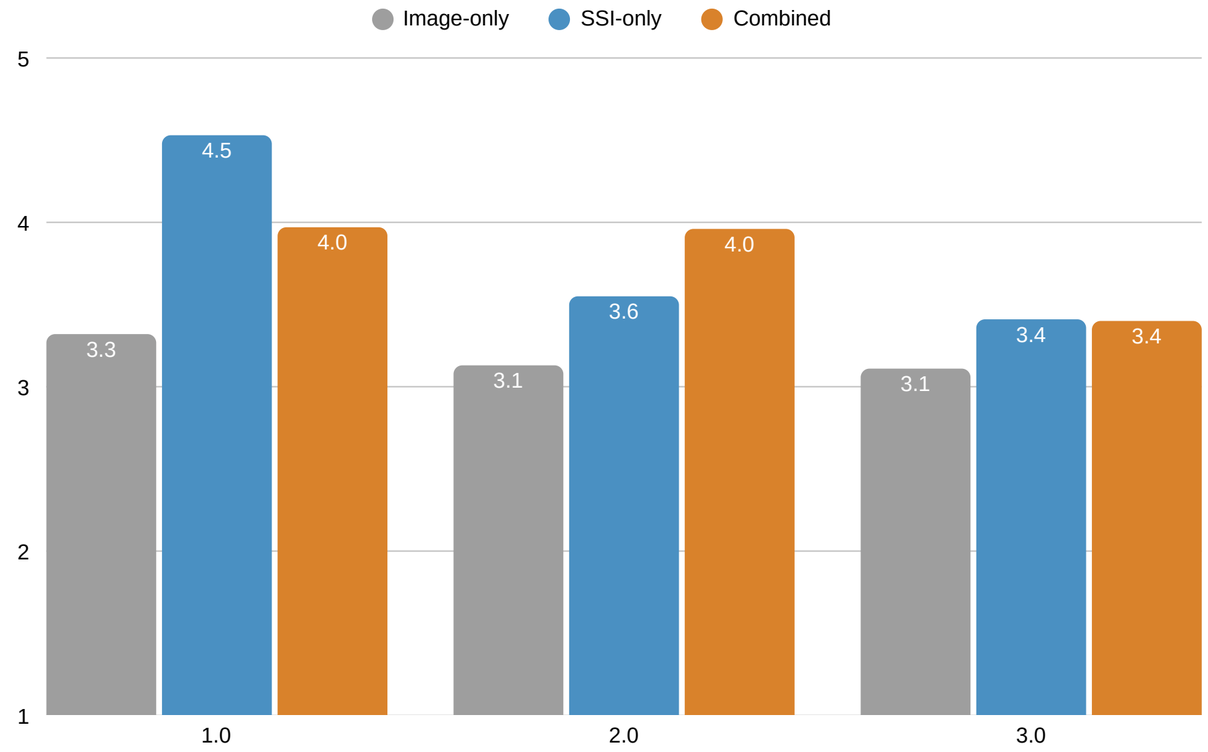
Safety note: The current significant rightward offset (31.5%) reduces the margin for error on both sides of the vehicle, increasing the risk of collisions with neighboring vehicles or roadside obstacles. Centering the vehicle will restore optimal safety margins.

→ **Select ising-calibration-31b:** stable, well-structured, higher score (|d_z| = 1.04); fixed for later experiments

# Evaluation of Recommendation Quality

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## E5: role of SSI (main result)




| Judge | Image-only | SSI-only | Combined | Δ | \|d_z\| |
|---|---|---|---|---|---|
| Gemini | 3.32 | 4.53 | 3.97 | **+0.65** | 0.59 |
| GPT-5 Mini | 3.13 | 3.55 | 3.96 | **+0.83** | 0.79 |
| DeepSeek | 3.11 | 3.41 | 3.40 | **+0.29** | 0.23 |

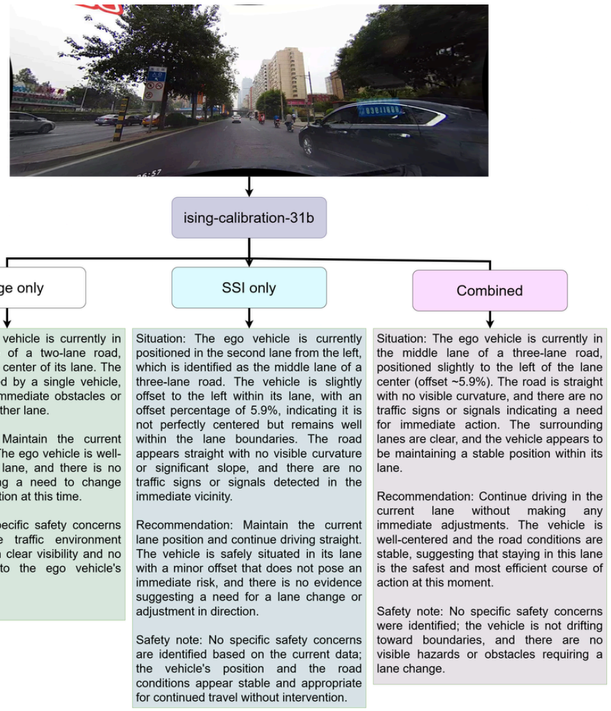
*Δ = Combined − Image-only*

→ **Image-only is lowest under all 3 judges**; differences significant (Bonferroni) · small-to-medium effects · SSI-only vs. Combined order inconsistent

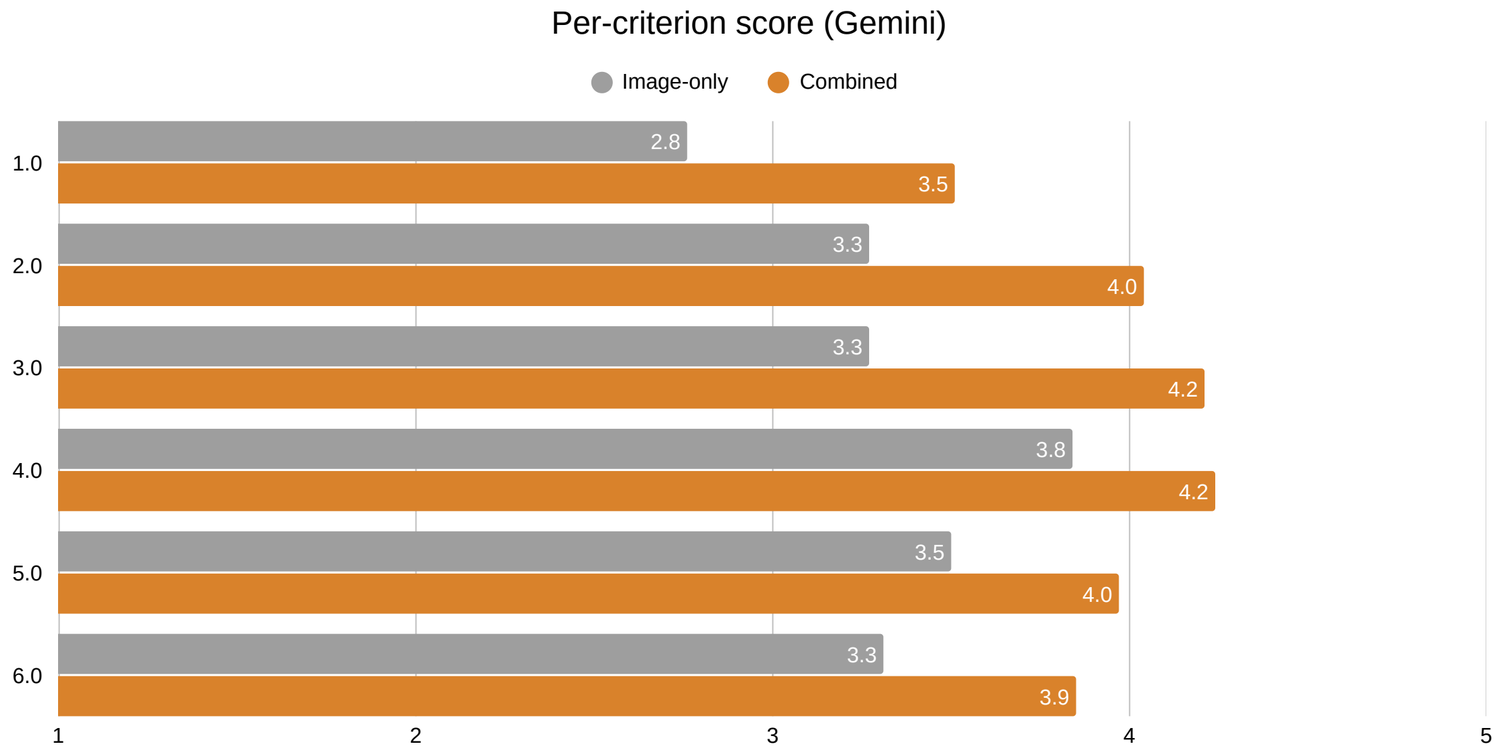
# Evaluation of Recommendation Quality

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## E5: which criteria SSI improves




✗ **Image-only:** "2 lanes, vehicle in the right lane"
✓ **With SSI:** "3 lanes, middle lane, offset 5.9%" → keep lane




→ Improvement on **all 6 criteria** ($p < 0.001$), including the driving recommendation and the safety note

# Evaluation of Recommendation Quality

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## E6: effect of SSI accuracy

**Split by ego lane in SSI:** correct 172 images · wrong 28 images

| Judge | Image-only (correct → wrong) | SSI-only (correct → wrong) | Combined (correct → wrong) |
|---|---|---|---|
| Gemini | 3.35 → 3.15 | **4.70 → 3.51** | 4.05 → 3.47 |
| GPT-5 Mini | 3.13 → 3.11 | **3.77 → 2.19** | **3.97 → 3.91** |
| DeepSeek | 3.17 → 2.75 | **3.55 → 2.50** | 3.47 → 2.95 |

→ **SSI-only** drops sharply when SSI is wrong ($p < 0.001$) · **Combined** never falls below Image-only in any group or judge

*The wrong-SSI group has only 28 images; results are indicative*

# Comparing LLM-as-a-Judge with Human Evaluation

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## E7: judge reliability

20 images · Combined mode · 120 score pairs per judge

| Judge | Exact agreement | Within ±1 | Pearson r (p) |
|---|---|---|---|
| **Gemini** | **36.7%** | **79.2%** | **0.473 (0.035)** |
| GPT-5 Mini | 23.3% | 62.5% | 0.358 (0.121) |
| DeepSeek | 25.8% | 55.0% | 0.054 (0.820) |

**79.2%**

Gemini scores within ±1 of humans

**r = 0.473**

the only significant correlation (p = 0.035)

→ **Gemini is closest to humans → primary judge** · per-criterion agreement is still low → used only for relative comparison between modes

# Supplementary Experiments

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## E8: VLM self-extraction of semantics



ising-calibration-31b


```
{
  "lane_count": 3,
  "ego_lane": {
    "position": "2/3",
    "confidence": 1.0
  },
  "vehicle_offset": {
    "direction": "centered",
    "magnitude": "none",
    "offset_percent": 0.0
  },
  "neighbor_lanes": {
    "left_count": 1,
    "right_count": 1
  },
  "road_shape": {
    "type": "straight",
    "severity": "none",
    "direction": null
  },
  "order": "left to right",
  "traffic_signs": {
    "detected": [],
    "count": 0
  }
}
```


| | Lane count (Pipeline / VLM) | Road shape (Pipeline / VLM) |
|---|---|---|
| CULane Normal | 77.3% / 54.5% | 84.1% / 95.5% |
| CULane Hard | 8.3% / 37.5% | 20.8% / 91.7% |
| Self-collected | 50.5% / 53.5% | 59.0% / 69.5% |

- No method wins on every field and every data group
- The VLM extracts semantics when asked directly, but does not exploit them when generating recommendations (Image-only: 3.27 on Road geometry and Ego-lane position)

→ SSI **complements** rather than replaces the VLM's perception

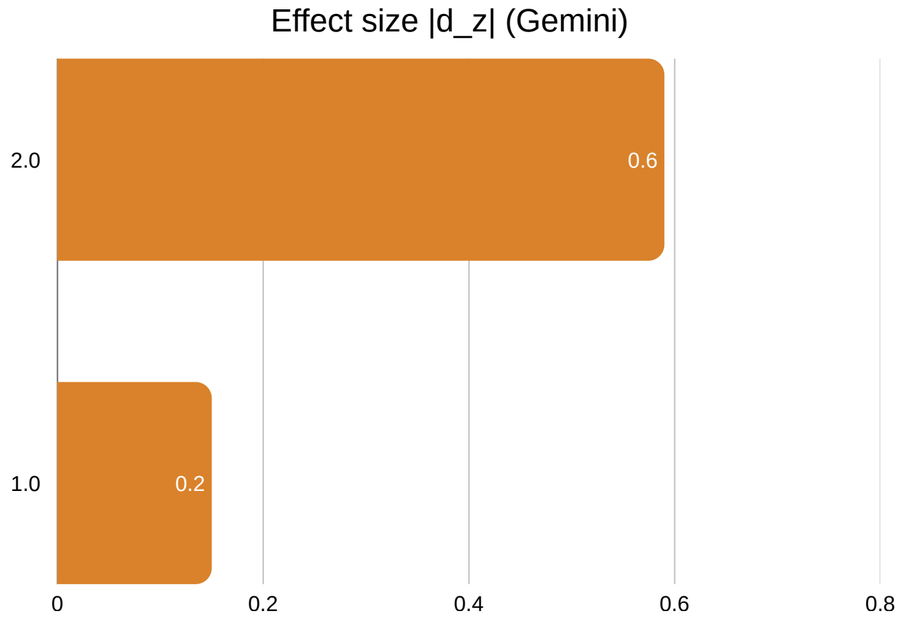
# Supplementary Experiments

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## E9: influence of the prompt

Same model, Combined mode, 200 images; only the prompt varies

| Prompt | Content | Score |
|---|---|---|
| v1 | Recommendation request only | 3.26 |
| v2 | v1 + 3-part output structure | 3.34 |
| **v3 (main)** | **v2 + evidence constraints, conflict handling** | **3.35** |




→ Prompt changes make only a **very small** difference → the main result is **not driven by the prompt**

FPT SCHOOL OF BUSINESS & TECHNOLOGY

# Conclusion

Answer to the research question · Limitations · Future work

# Conclusion

FPT SCHOOL OF BUSINESS & TECHNOLOGY

**YES**

**Does adding SSI improve the VLM's recommendations?**
Combined > Image-only under 3/3 judges, Δ = +0.29 to +0.83 / 5, $p < 0.001$

| Objective | Evidence |
|---|---|
| ① SSI extraction ✓ | Ego lane ~86% on 2 datasets · 0.55 s/image on CPU |
| ② SSI-based recommendation ✓ | Stable on 200/200 images, correct structure · Combined ≥ Image-only even when SSI is wrong |
| ③ Role of SSI evaluated ✓ | Gains on all 6 criteria, consistent across 3 judges · not due to the prompt · Gemini closest to humans |

# Limitations and Discussion

| Limitation | Mitigation |
| --- | --- |
| SSI depends on detection quality (Hard images, signs outside the 50 classes) | Combined mode + conflict-resolution rules |
| 200 CULane images not from the test split; possible overlap with training data | 200 additional out-of-domain self-collected images |
| Per-criterion judge–human agreement still low | 3 judges, human comparison, relative comparison only |
| Not yet real-time; API cost not evaluated | SSI runs on CPU; latency is mostly the VLM |

# Future Work

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## 1 Improving SSI reliability

- Better detection models
- SSI checking and correction
- Per-field SSI ablation

## 2 Optimization and extension

- Lower latency
- Lightweight / on-device VLMs
- Vietnamese traffic data, more diverse conditions

FPT SCHOOL OF BUSINESS & TECHNOLOGY

# Thank you for your attention

*SSI: a lightweight semantic layer that helps VLMs give better driving recommendations, and keeps quality when SSI is wrong*

# Semantic Representation

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## [Backup] Ego-lane penalty and confidence levels

Penalty of the candidate lane formed by boundaries (b_i, b_i+1):

$$s_i = \left| \frac{(x_i + x_{i+1})}{2} - x_v \right| \cdot \beta_i \cdot \left( 1 + 0.2 \cdot \frac{|w_i - w_0|}{w_0} \right)$$

- $w_i = x_{i+1} - x_i$, $w_0 = 0.15W$, $x_v = W/2 + o_c$ ($o_c = 0$ without calibration)
- $\beta_i = 0.5$ if $x_v$ lies inside the lane, otherwise 1; choose the lane with the smallest $s_i$
- Confidence 0.95 / 0.8 / 0.6 / 0.4 when the lane-center distance is $< 0.1W$ / $0.25W$ / $0.4W$ / otherwise
- Only one boundary: treated as the ego lane's right boundary, confidence 0.5; no boundary: empty

# Semantic Representation

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## [Backup] Curvature and road-shape classification

- Each boundary with ≥ 4 points: y normalized by image height; fit 1st- and 2nd-order polynomials ($MSE_1$, $MSE_2$)
- Deviation from a straight line: $\delta = \sqrt{MSE_1} / W$
- 2nd-order gain: $\eta = 0$ if $MSE_1 \le 4$; else $\eta = \max(0, (MSE_1 - MSE_2) / MSE_1)$ (only with ≥ 10 points)
- Normalized lateral shift: $\sigma = 1/(W \cdot |B'|) \cdot \Sigma (\bar{x}_bot - \bar{x}_top)$, means of the 20% points nearest the bottom / top
- A boundary is straight if $MSE_1 < 1000\ px^2$; r_straight = share of straight boundaries
- Straight if $\bar{\delta} < 0.02 \wedge \bar{\eta} < 0.3 \wedge$ r_straight > 0.5; sharp if $\bar{\delta} \ge 0.06$; otherwise slight
- Direction from the sign of σ: σ > 0 right, σ < 0 left

*See Eqs. (3.11)–(3.14) in the thesis*

# Semantic Representation

FPT SCHOOL OF BUSINESS & TECHNOLOGY

[Backup] Heuristic parameters and how they were chosen

- Penalty coefficient 0.2 and standard lane width $w_0 = 0.15W$
- Confidence levels 0.95 / 0.8 / 0.6 / 0.4
- 10 px threshold for "centered"; 10% and 30% for offset magnitude
- 20% of points near bottom/top; minimum boundary length 5% of image height
- $MSE_1 < 1000$ for a straight boundary; thresholds on $\delta$, $\eta$, $r_{straight}$
- $\rho = 0.919$: visible image fraction above the hood in CULane (camera mounting; no ground truth used)

→ Chosen manually on sample images to discretize geometry into a few VLM-friendly levels. **Not tuned on ground truth.** No sensitivity analysis yet (a stated limitation).

# Model and Data Selection

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## [Backup] The 50 TT100K sign classes

| Group | Class codes | Meaning |
|---|---|---|
| Prohibitory | p1, p3, p5, p6, p10, p11, p12, p13, p19, p23, p26, p27, pb, pbp, pn, pne | No straight ahead, no buses, no U-turn, no bicycles, no motor vehicles, no honking, no motorcycles, no tractors, no right turn, no left turn, no trucks, no dangerous goods, no entry / wrong way, no tricycles, no parking, no entry |
| | pl5, pl10, pl15, pl20, pl25, pl30, pl35, pl40, pl50, pl60, pl65, pl70, pl80, pl90, pl100, pl110, pl120 | Maximum speed N km/h |
| | pm, pa, ph, pr | Weight limit 10 t, axle load 10 t, height 3.5 m, end of 40 km/h limit |
| Mandatory | pg, i2, i4, i5 | Yield, bicycle lane, motor-vehicle lane, mandatory direction |
| Guide | ip; il50, il60, il70, il80, il90, il100, il110 | Pedestrian and bicycle path; recommended speed N km/h |
| Warning | w | Warning sign (generic) |

# Evaluation of Recommendation Quality

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## [Backup] Full Wilcoxon tests for the 9 mode pairs

| Judge | Pair | Wilcoxon p | Cohen's d_z |
|---|---|---|---|
| Gemini | Image-only – SSI-only | < 0.001 | −0.96 |
| Gemini | Image-only – Combined | < 0.001 | −0.59 |
| Gemini | SSI-only – Combined | < 0.001 | 0.51 |
| GPT-5 Mini | Image-only – SSI-only | < 0.001 | −0.31 |
| GPT-5 Mini | Image-only – Combined | < 0.001 | −0.79 |
| GPT-5 Mini | SSI-only – Combined | < 0.001 | −0.35 |
| DeepSeek | Image-only – SSI-only | 0.003 | −0.20 |
| DeepSeek | Image-only – Combined | < 0.001 | −0.23 |
| DeepSeek | SSI-only – Combined | 0.625 | 0.01 |

*Bonferroni threshold 0.0056. Negative d_z: the first mode of the pair scores lower*

# Supplementary Experiments

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## [Backup] 6 criteria across the 3 prompt versions

| Criterion | Prompt v1 | Prompt v2 | Prompt v3 |
|---|---|---|---|
| Situation | 3.11 ± 0.84 | 3.11 ± 0.86 | **3.15 ± 0.76** |
| Road geometry | 3.50 ± 0.68 | 3.51 ± 0.66 | **3.52 ± 0.64** |
| Ego-lane position | 3.52 ± 0.80 | 3.58 ± 0.75 | **3.62 ± 0.71** |
| Signs and rules | 3.10 ± 0.67 | 3.13 ± 0.66 | **3.23 ± 0.79** |
| Driving recommendation | 3.29 ± 0.84 | 3.42 ± 0.78 | **3.52 ± 0.74** |
| Safety note | 3.03 ± 0.72 | **3.27 ± 0.75** | 3.05 ± 0.76 |
| **Mean of 6 criteria** | 3.26 ± 0.53 | 3.34 ± 0.50 | **3.35 ± 0.47** |

*v2 vs. v1: p = 0.012; v3 vs. v1: p = 0.028 (above the 0.0167 threshold); v3 vs. v2: p = 0.77*

# Experimental Setup

FPT SCHOOL OF BUSINESS & TECHNOLOGY

## [Backup] Judge and VLM configuration

| Judge | Model | Configuration |
|---|---|---|
| Gemini | gemini-3.5-flash | temperature 0.1 · max 8192 tokens |
| GPT-5 Mini | gpt-5-mini | max 4096 tokens · default temperature |
| DeepSeek | deepseek-v4-flash-vision-exp | temperature 0.1 · max 8192 tokens |

- **Judges:** up to 5 retries on transient errors
- **VLM:** max_tokens 400 · temperature 0.2 · top_p 0.7 · frequency_penalty 0.4 · timeout 120 s · image ≤ 150 KB